E-Cycle Connect: Project Design Document

Team LOGICLOOP | Technology and Sustainability (2026CSAE102) | Milestone 2

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1. Overview

Problem. Old electronics are often thrown away or passed to informal scrap channels because people do not know whether a device can be repaired, reused or must be recycled, or where to take it safely.

Solution. E-Cycle Connect is a digital platform where a user registers an old device, receives a repair, reuse or recycle recommendation, books a drop-off at a nearby partner, and tracks the device until its journey ends.

SDG link. The strongest link is SDG 12 (Responsible Consumption and Production). The platform also supports SDG 9 (digital tracking infrastructure), SDG 11 (safer e-waste handling in communities) and SDG 13 (less unnecessary production and disposal).

2. Users and Requirements

Users: students, households, schools and offices (device owners); repair shops, recyclers and donation drives (partners); a verifier who approves partners (planned).

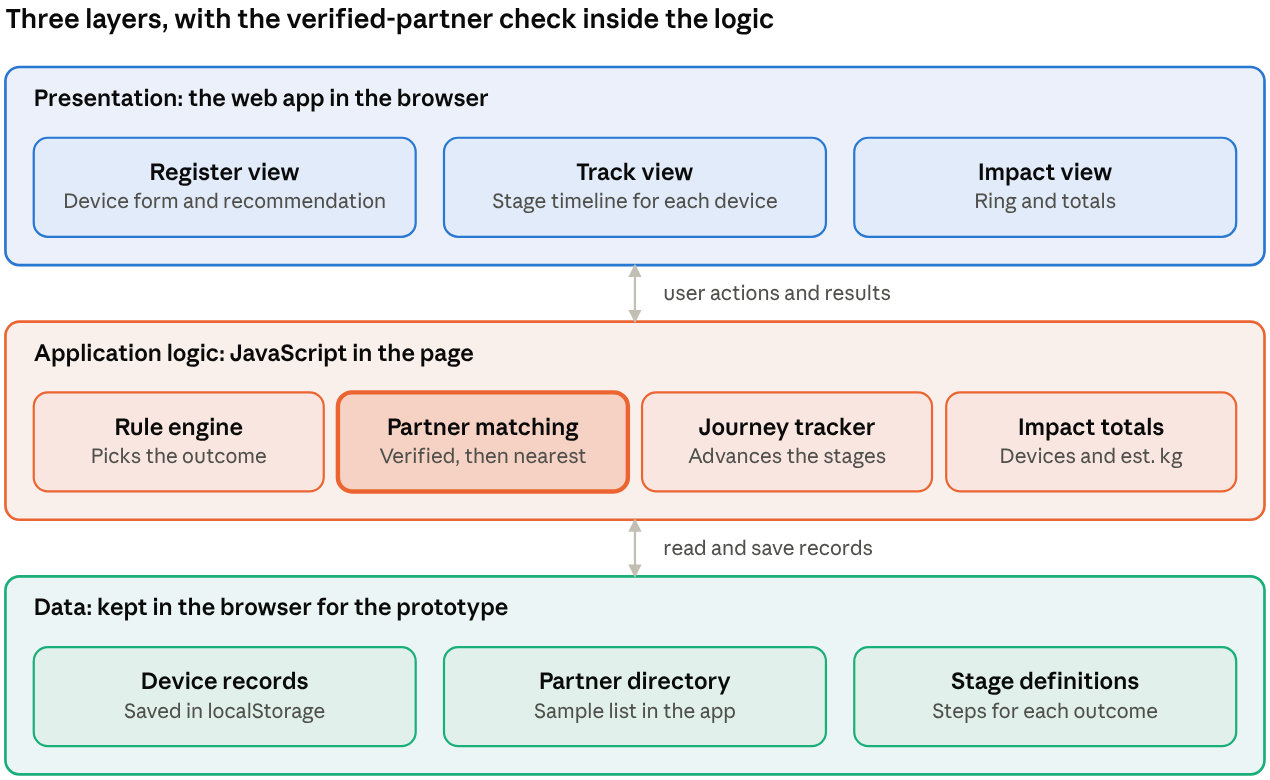
Functional requirements

- FR1: Register a device with its type, model, age and condition.
- FR2: Recommend reuse, repair or recycle using clear rules.
- FR3: List nearby partners that accept the recommended outcome, with their verification status.
- FR4: Block recycling through unverified partners.
- FR5: Book a drop-off and track the device through its stages.
- FR6: Show an impact summary (devices handled, estimated e-waste kept out of landfill).

Non-functional requirements: simple enough to use in under a minute, works on a phone screen, needs no sign-up in the prototype, stores no personal data, and supports keyboard use, dark mode and reduced motion.

3. System Architecture

The system has three layers. The prototype runs entirely in the browser, and the planned production version separates the layers across a server.



system architecture · 3 layers, 10 components

Figure 1. The web app sends user actions to the logic layer, which reads and saves records in the data layer. The partner check sits in the logic layer, so recycling can never reach an unverified partner.

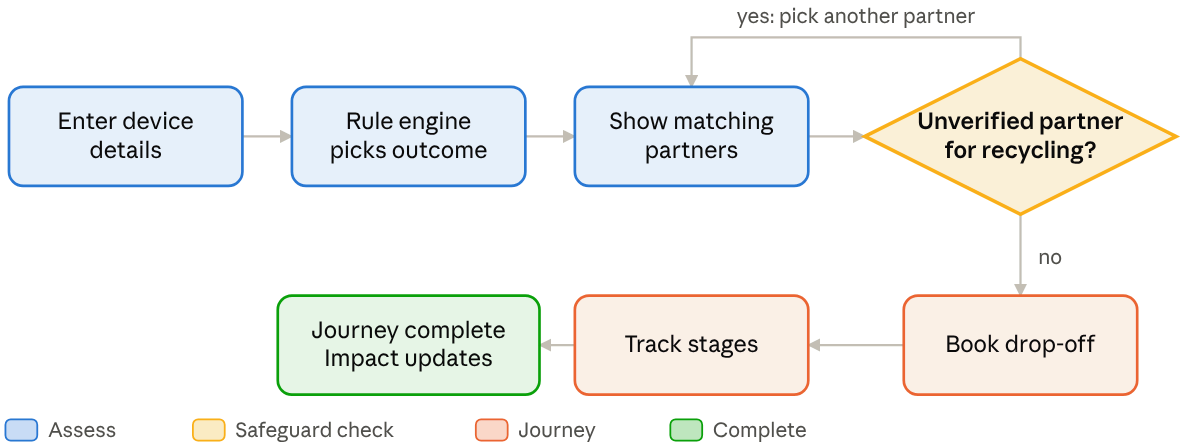
Layer	Prototype (built)	Planned production
Presentation	Single-page web app with Register, Track and Impact views	Responsive web app and mobile app

Layer	Prototype (built)	Planned production
Application logic	Recommendation engine, partner matching with verification filter, journey tracker, impact calculator (JavaScript)	Same logic as a REST API, plus a partner portal for status updates
Data	Device records in browser localStorage, sample partners hard-coded	Database for devices, partners and tracking events, plus a map service for distances

4. Workflow

1. The user opens the app and enters the device type, model, age and condition.
2. The recommendation engine applies its rules and returns reuse, repair or recycle with a short reason.
3. The app lists partners that accept that outcome, verified partners first, then nearest.
4. If the outcome is recycle, unverified partners are disabled.
5. The user books a drop-off. A device record is created at stage 1.
6. The device moves through the stages for its outcome as the partner updates its status. In the prototype, the Advance stage button simulates this.
7. When the last stage is reached, the journey is complete and the impact summary updates.

Only verified partners can take recycling



workflow · 7 steps, 1 decision

Figure 2. Every device gets one outcome from the rule engine. If the outcome is recycle and the partner is unverified, the app does not allow the booking and the user picks another partner.

5. Data Flow

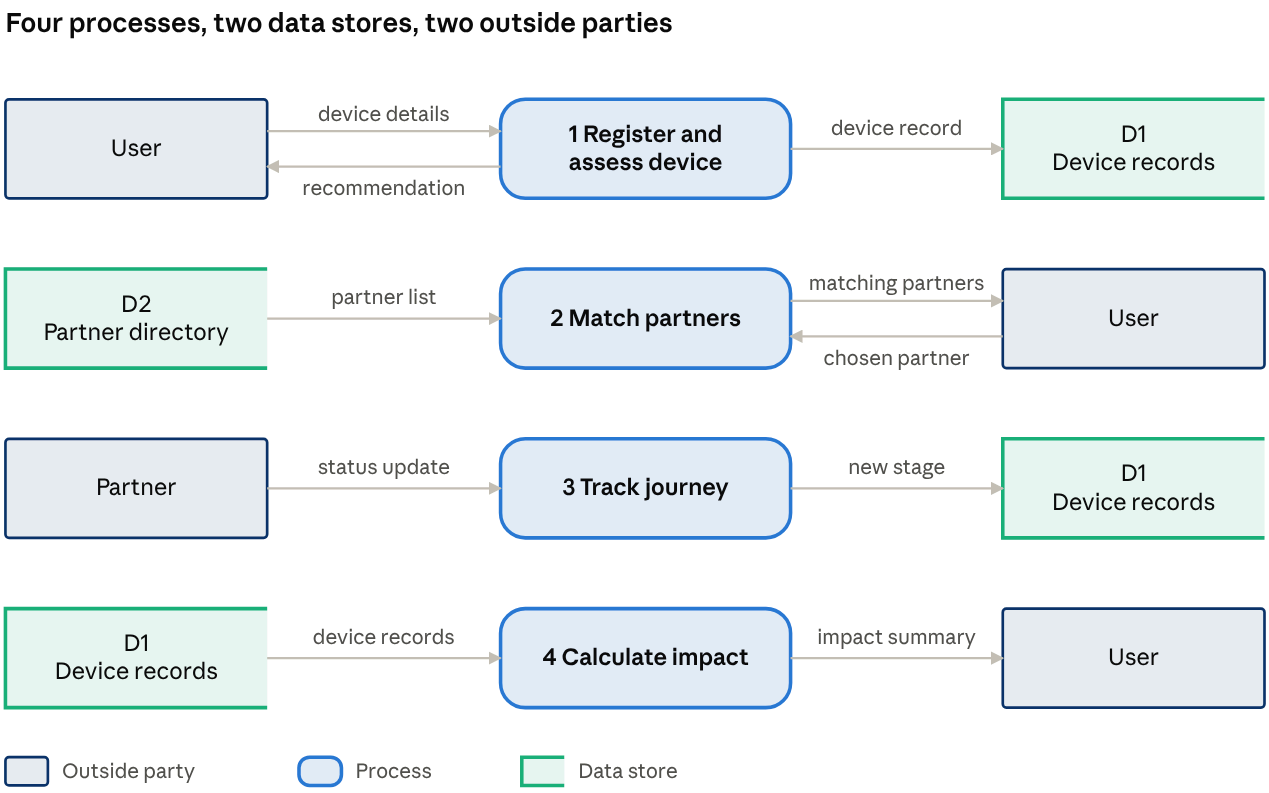
External entities: User, Partner.

Processes: (1) Register and assess device, (2) Match partners, (3) Track journey, (4) Calculate impact.

Data stores: D1 Device records, D2 Partner directory.

Main flows

- User sends device details to Process 1. Process 1 saves the record in D1 and returns a recommendation.
- Process 2 reads D2 using the recommended outcome and returns matching partners to the User.
- The User's chosen partner is saved against the device record in D1.
- Partner sends status updates to Process 3, which updates the stage in D1.
- Process 4 reads D1 and returns the impact summary to the User.



data flow diagram · 4 processes, 2 stores

Figure 3. Booking a drop-off also saves the chosen partner on the device record in D1.

6. Data Model

Device record (stored as one object per device)

Field	Meaning
id	Unique number (timestamp)
type	Phone, Laptop, Charger, Battery, TV / monitor or Other
model	Brand and model typed by the user
a	Outcome: reuse, repair or recycle
w	Estimated weight in kg, based on device type
point	Name of the chosen partner

Field	Meaning
stage	Current stage number in the journey

Partner record: name, distance in km, verified (true or false), list of outcomes accepted.

Stages by outcome

Outcome	Stages
Reuse	Registered, Handed over, Quality checked, Reused or donated
Repair	Registered, Handed over, Repaired, Returned or reused
Recycle	Registered, Collected by verified recycler, Dismantled and sorted, Materials recovered

7. Recommendation Rules

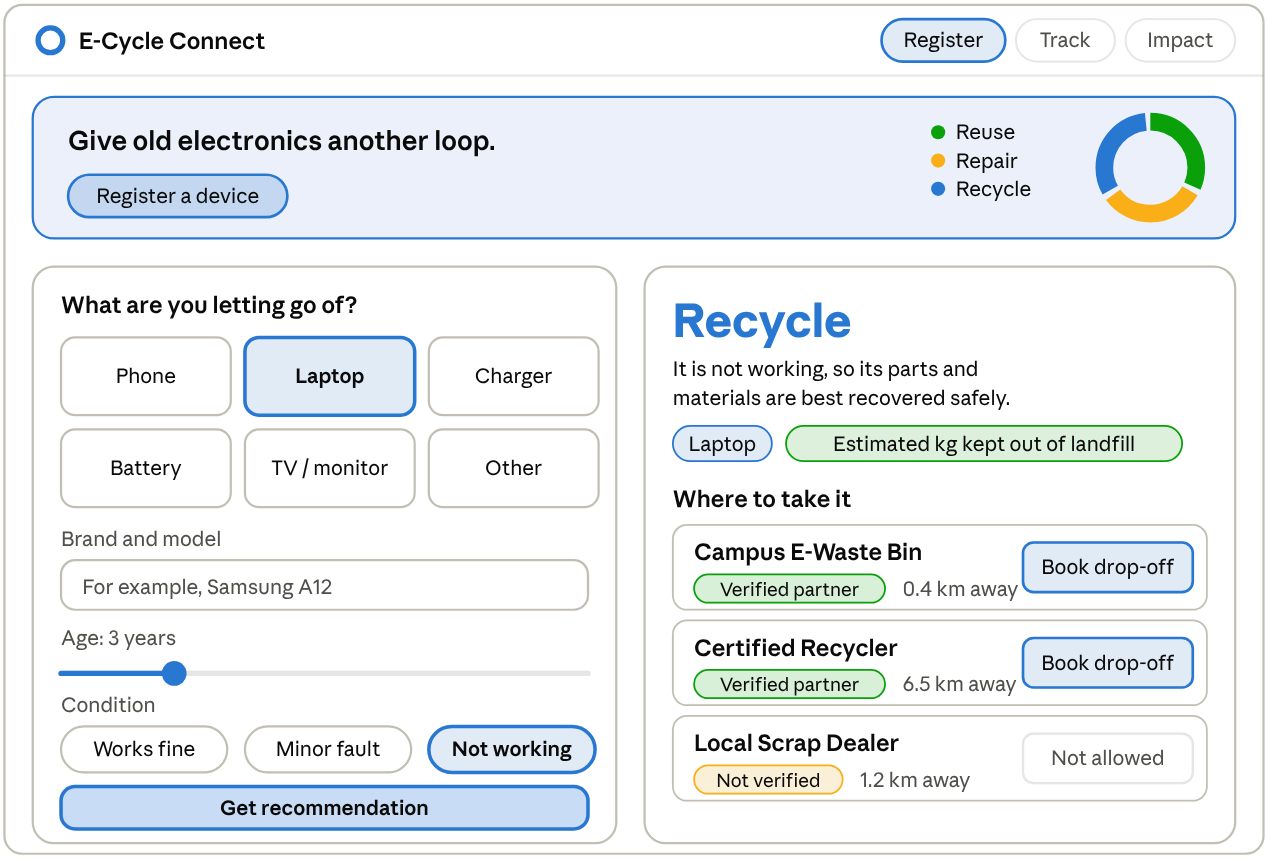
Input	Outcome	Reason shown
Device type is Battery	Recycle	Batteries are hazardous and go only to a verified recycler
Works fine	Reuse	Donating or reselling keeps it in use the longest
Minor fault	Repair	Usually cheaper to repair than to replace
Not working	Recycle	Parts and materials are recovered safely

The battery rule is checked first. The verification rule then applies to every recycle outcome: only verified partners can be booked.

8. User Interface Design

The interface has three views, switched from the top bar. A hero section with a loop ring shows how many devices have been registered and how they split between reuse, repair and recycle.

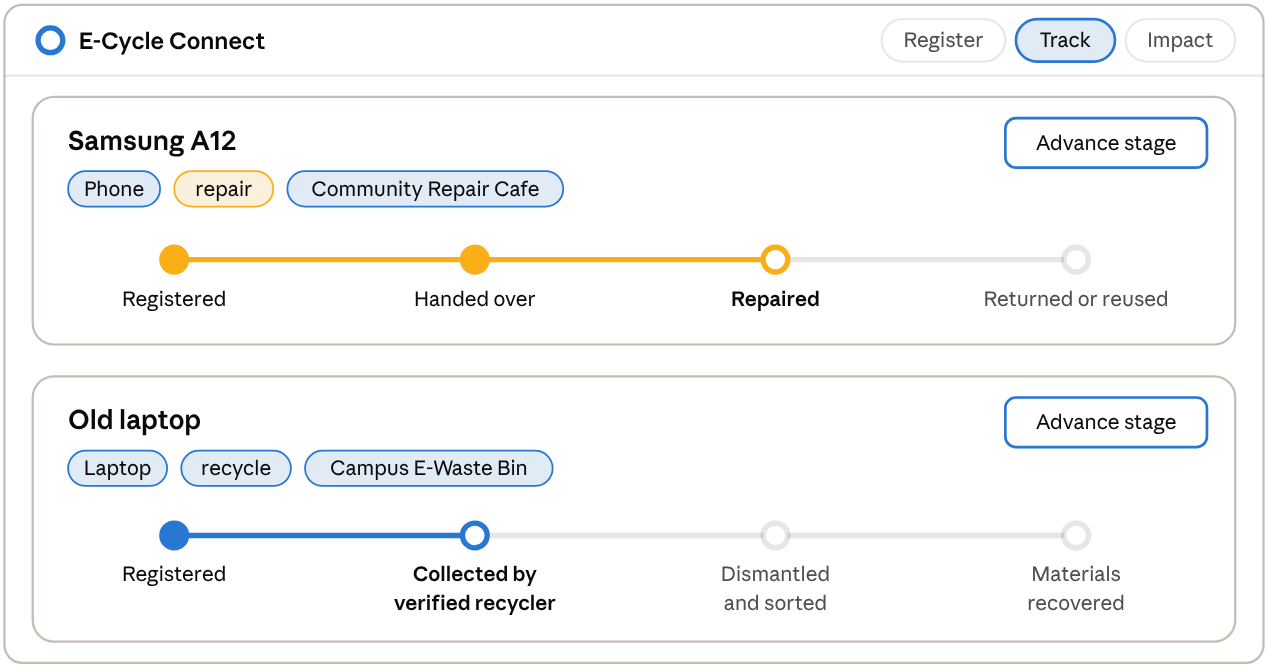
Register view



Register screen wireframe

The Register screen. The Not allowed button shows the safeguard: unverified partners cannot take recycling.

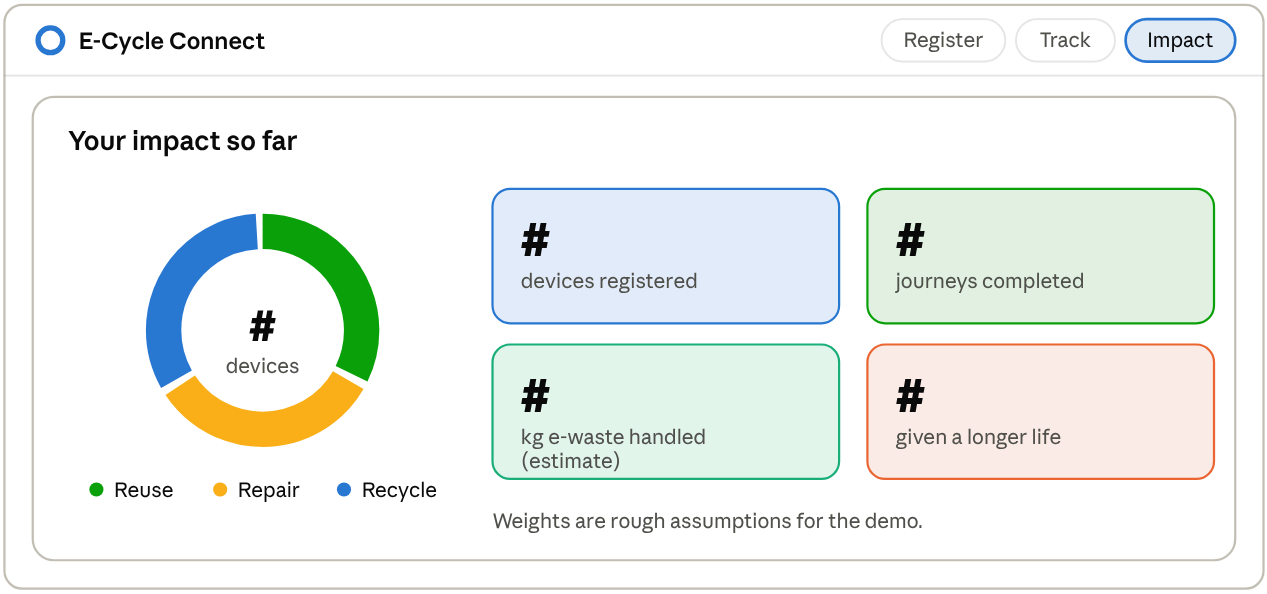
Track view



Track screen wireframe

The Track screen. Each device card shows its journey in the colour of its outcome.

Impact view



Impact screen wireframe

The Impact screen. The ring splits devices into reuse, repair and recycle.

9. Prototype Status

Built and working: device registration, rule-based recommendation, partner list with the verification rule, drop-off booking, stage tracking, impact summary, saved data between visits, responsive layout and dark mode.

Not yet built: accounts and sign-in, real partner data and live map distances, partner portal for real status updates, and scanning a device by camera or barcode. The prototype uses manual entry instead of scanning.

10. Risks and Safeguards

Risk	Safeguard in the design
Poorly managed recycling partners could shift the burden elsewhere	Only verified partners can receive recycling; unverified ones are disabled
Users need devices, electricity and internet to take part	Simple, light interface; partner drop-offs on campus and in the neighbourhood
Hazardous items such as batteries handled unsafely	Battery rule forces recycle through a verified recycler
Estimates may be mistaken for exact figures	Weights are labelled as estimates throughout

11. Next Steps (Milestone 3)

- Replace sample partners with real partners and add a verification process.
- Add a backend and shared database so partners can update stages.
- Test with students and one repair or recycling partner, and record their feedback.
- Finalise the report, presentation and demonstration.